

ПУНКТ 1

Для построения топологии без петель коммутации был настроен протокол классического Spanning Tree. Чтобы сделать выбор топологии детерминированным, один из коммутаторов (Layer2Switch-1) был принудительно назначен корневым путем снижения его приоритета Spanning Tree для VLAN 1 до значения 4096. Коммутатор Layer2Switch-2 был назначен резервным корневым коммутатором с приоритетом 8192.

Команды настройки основного корневого коммутатора (Layer2Switch-1):

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#spanning-tree vlan 1 prio
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#end
*Sep  7 13:24:11.640: %SYS-5-CONFIG_I: Configured from console by consoley
vIOS-L2-01#write memory
Building configuration...
```

Команды настройки резервного корневого коммутатора (Layer2Switch-2):

```
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#spanning-tree vlan 1 priority 8192
vIOS-L2-01(config)#end
*Sep  7 13:26:33.520: %SYS-5-CONFIG_I: Configured from console by consolememory
vIOS-L2-01#write memory
Building configuration...
```

Команды настройки коммутаторов доступа (Layer2Switch-3/4/5):

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#write me
*Sep  7 13:27:28.244: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
```

ПУНКТ 2

Для проверки доступности узлов на персональных компьютерах были настроены IP-адреса из единой подсети 192.168.1.0/24.

Команды настройки IP на хостах:

```
PC1> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

```
PC2> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0
```

```
PC3> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0
```

```
PC4> ip 192.168.1.4 255.255.255.0
Checking for duplicate address...
PC4 : 192.168.1.4 255.255.255.0
```

```
PC5> ip 192.168.1.5 255.255.255.0
Checking for duplicate address...
PC5 : 192.168.1.5 255.255.255.0
```

```
PC6> ip 192.168.1.6 255.255.255.0
Checking for duplicate address...
PC6 : 192.168.1.6 255.255.255.0
```

Проверка доступности всех остальных компьютеров с PC1:

```
PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=49.597 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=49.824 ms
^C
PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=86.247 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=47.917 ms
^C
PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=52.036 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=27.030 ms
```

^C

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=36.039 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=64.480 ms

^C

PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=11.282 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=12.875 ms

^C

PC1>

Результаты взаимного опроса (ping) в следующей таблице связности:

	PC1	PC2	PC3	PC4	PC5	PC6
PC1	*	+	+	+	+	+
PC2	+	*	+	+	+	+
PC3	+	+	*	+	+	+
PC4	+	+	+	*	+	+
PC5	+	+	+	+	*	+
PC6	+	+	+	+	+	*

ПУНКТ 3

Выполнение команды show spanning-tree vlan 1 для каждого коммутатора по возрастанию (Layer2Switch-1 -> Layer2Switch-2 -> ...):

```
vIOS-L2-01>show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097
---------	----------	------

Address 0cef.e827.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cef.e827.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg FWD 4	128.1	Shr
-------	------------	-------	-----

Gi0/1	Desg FWD 4	128.2	Shr
-------	------------	-------	-----

Gi0/2	Desg FWD 4	128.3	Shr
-------	------------	-------	-----

Gi0/3	Desg FWD 4	128.4	Shr
-------	------------	-------	-----

Gi1/0	Desg FWD 4	128.5	Shr
-------	------------	-------	-----

Gi1/1	Desg FWD 4	128.6	Shr
-------	------------	-------	-----

Gi1/2	Desg FWD 4	128.7	Shr
-------	------------	-------	-----

Gi1/3	Desg FWD 4	128.8	Shr
-------	------------	-------	-----

```
vIOS-L2-01>show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097
---------	----------	------

Address 0cef.e827.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 8193 (priority 8192 sys-id-ext 1)

Address 0ca9.cb37.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root FWD 4	128.1	Shr
Gi0/1	Altn BLK 4	128.2	Shr
Gi0/2	Desg FWD 4	128.3	Shr
Gi0/3	Desg FWD 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr
Gi1/2	Desg FWD 4	128.7	Shr
Gi1/3	Desg FWD 4	128.8	Shr
Gi2/0	Desg FWD 4	128.9	Shr

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cef.e827.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c85.9142.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root FWD 4		128.1			Shr
Gi0/1	Altn BLK 4		128.2			Shr
Gi0/2	Altn BLK 4		128.3			Shr
Gi0/3	Altn BLK 4		128.4			Shr
Gi1/0	Desg FWD 4		128.5			Shr
Gi1/1	Desg FWD 4		128.6			Shr

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cef.e827.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c37.766b.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec				
Interface	Role	Sts	Cost	Prio.Nbr Type

Gi0/0	Root	FWD	4	128.1 Shr
Gi0/1	Altn	BLK	4	128.2 Shr
Gi0/2	Altn	BLK	4	128.3 Shr
Gi0/3	Altn	BLK	4	128.4 Shr
Gi1/0	Desg	FWD	4	128.5 Shr
Gi1/1	Desg	FWD	4	128.6 Shr

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cef.e827.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c08.781a.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr Type

Gi0/0	Root	FWD	4	128.1 Shr
Gi0/1	Altn	BLK	4	128.2 Shr
Gi0/2	Altn	BLK	4	128.3 Shr
Gi0/3	Altn	BLK	4	128.4 Shr
Gi1/0	Desg	FWD	4	128.5 Shr
Gi1/1	Desg	FWD	4	128.6 Shr

По результатам выполнения команды show spanning-tree vlan 1 были определены параметры Spanning Tree для каждого коммутатора:

1. Корневой коммутатор (Layer2Switch-1):

BID: 4097 / 0cef.e827.0000 (Приоритет 4096 + VLAN ID 1).

Роли портов: Все активные порты находятся в состоянии **Desg FWD** (Designated, Forwarding) со стоимостью пути до корня Cost = 0

2. Резервный корневой коммутатор (Layer2Switch-2):

BID: 8193 / 0ca9.cb37.0000 (Приоритет 8192 + VLAN ID 1)

Роли портов: Порт в сторону Layer2Switch-1 имеет роль **Root FWD** (Cost = 4). Порты в сторону коммутаторов доступа - **Desg FWD**.

3. Коммутатор доступа vIOS-L2-01 (Layer2Switch-3):

BID: 32769 / 0c85.9142.0000 (Приоритет 32768 + VLAN ID 1)

Параметры портов:

Gi0/0 - **Root FWD** (Корневой порт), стоимость Cost = 4

Gi0/1, Gi0/2, Gi0/3 - **Altn BLK** (Альтернативные заблокированные порты), стоимость Cost = 4

Gi1/0, Gi1/1 - **Desg FWD** (Назначенные порты к компьютерам PC1, PC2), стоимость Cost = 4

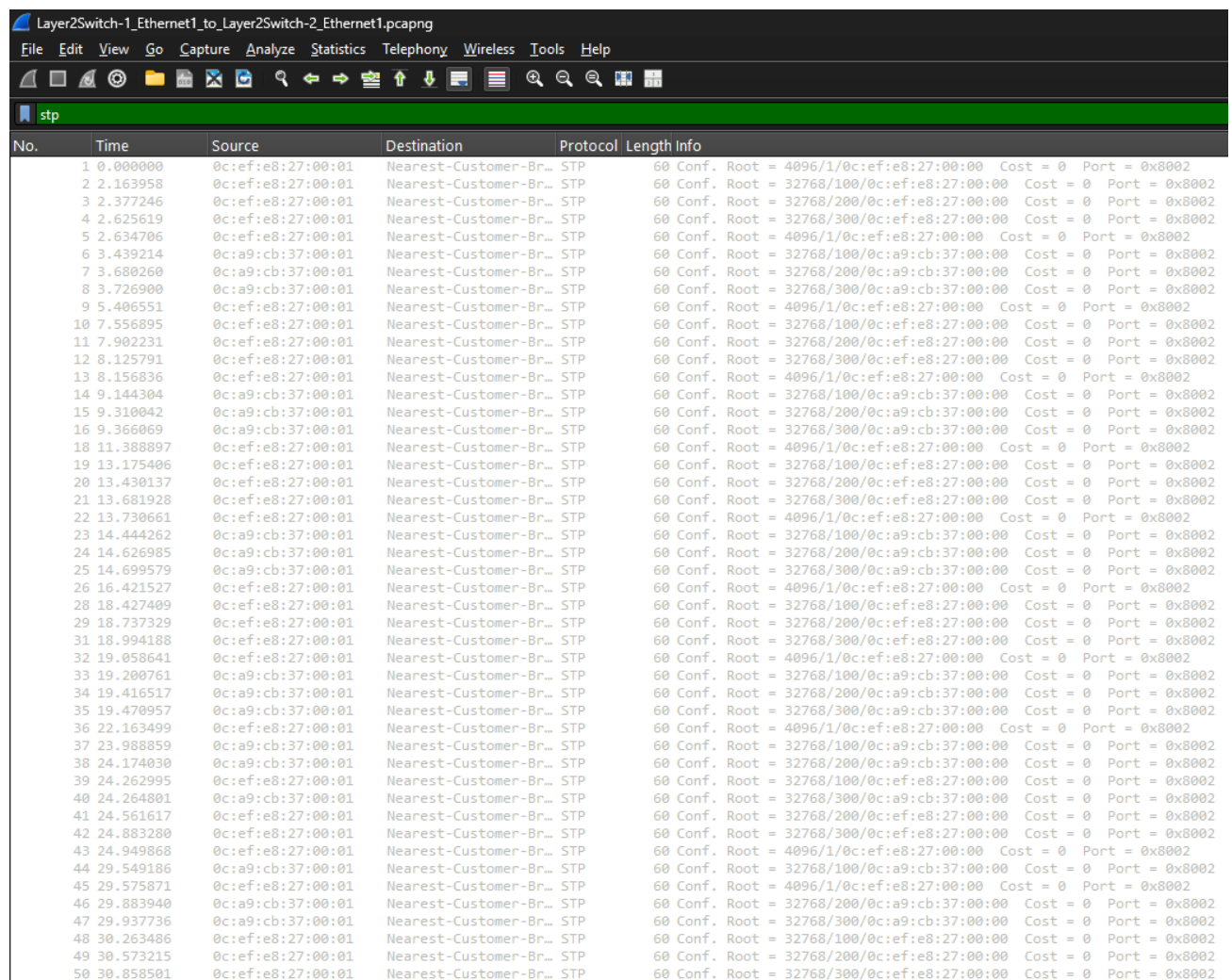
ПУНКТ 4

Для отслеживания передачи пакетов Hello (BPDU) было выполнено 4 выборочных захвата трафика на ключевых направлениях

Захват №1: Layer2Switch-1 (Root, Eth1) <-> Layer2Switch-2 (Backup, Eth1)

Зафиксирована работа протокола PVST+. Hello-пакеты отправляются отдельно для VLAN 1(1 пакет), VLAN 100(2 пакет), VLAN 200(3 пакет), VLAN 300(4 пакет) каждые 2 секунды

Для VLAN 1 в пакетах указано: Conf. Root = 4096/1/0cef.e827.0000, Cost = 0. Это подтверждает, что первоисточником является Layer2Switch-1



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
2	2.163958	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
3	2.377246	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
4	2.625619	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
5	2.634706	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
6	3.439214	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
7	3.680260	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
8	3.726900	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
9	5.406551	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
10	7.556895	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
11	7.902231	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
12	8.125791	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
13	8.156836	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
14	9.144304	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
15	9.310042	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
16	9.366069	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
18	11.388897	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
19	13.175406	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
20	13.430137	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
21	13.681928	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
22	13.730661	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
23	14.444262	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
24	14.626985	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
25	14.699579	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
26	16.421527	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
28	18.427409	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
29	18.737329	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
31	18.994188	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
32	19.058641	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
33	19.200761	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
34	19.416517	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
35	19.470957	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
36	22.163499	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
37	23.988859	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
38	24.174030	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
39	24.262995	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
40	24.264801	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
41	24.561617	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
42	24.883280	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
43	24.949868	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
44	29.549186	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
45	29.575781	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
46	29.883940	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
47	29.937736	0c:a9:cb:37:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8002
48	30.263486	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
49	30.573215	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002
50	30.858501	0c:ef:e8:27:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8002

Первый, второй, третий и четвертый пакет

```
▶ Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ IEEE 802.3 Ethernet
▶ Logical-Link Control
▼ Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  ▶ BPDU flags: 0x00
  ▶ Root Identifier: 4096 / 1 / 0c:ef:e8:27:00:00
    Root Path Cost: 0
  ▶ Bridge Identifier: 4096 / 1 / 0c:ef:e8:27:00:00
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

```
▶ Frame 2: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ Ethernet II, Src: 0c:ef:e8:27:00:01 (0c:ef:e8:27:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
▶ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
▶ Logical-Link Control
▼ Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  ▶ BPDU flags: 0x00
  ▶ Root Identifier: 32768 / 100 / 0c:ef:e8:27:00:00
    Root Path Cost: 0
  ▶ Bridge Identifier: 32768 / 100 / 0c:ef:e8:27:00:00
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

```
▶ Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ Ethernet II, Src: 0c:ef:e8:27:00:01 (0c:ef:e8:27:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
▶ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
▶ Logical-Link Control
▼ Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  ▶ BPDU flags: 0x00
  ▶ Root Identifier: 32768 / 200 / 0c:ef:e8:27:00:00
    Root Path Cost: 0
  ▶ Bridge Identifier: 32768 / 200 / 0c:ef:e8:27:00:00
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

```
▶ Frame 4: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ Ethernet II, Src: 0c:ef:e8:27:00:01 (0c:ef:e8:27:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
▶ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300
▶ Logical-Link Control
▼ Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  ▶ BPDU flags: 0x00
  ▶ Root Identifier: 32768 / 300 / 0c:ef:e8:27:00:00
    Root Path Cost: 0
  ▶ Bridge Identifier: 32768 / 300 / 0c:ef:e8:27:00:00
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

Захват №2: Layer2Switch-1 (Root, Eth2) <-> Layer2Switch-3 (Access, Eth0)

Есть отправка BPDU от корня (0cef.e827.0002) со стоимостью Cost = 0 к коммутатору доступа Layer2Switch-3 (0c85.9142.0000) для VLAN 1, 100, 200, 300

Порт Gi0/0 на Layer2Switch-3 принимает эти сообщения и переходит в статус Root FWD

stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
2	0.305034	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
3	0.308561	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
5	0.439603	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
6	0.578907	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
7	0.639980	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
8	0.646469	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
9	2.871770	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
10	4.426102	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
11	4.813975	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
12	4.821246	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
13	5.588358	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
14	5.829388	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
15	5.927666	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
16	6.166527	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
18	8.542189	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
19	8.838374	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
20	9.216804	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
21	9.225652	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
22	10.981207	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
23	11.256137	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
24	11.562366	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
25	11.779860	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
26	13.139277	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
27	13.451286	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
28	13.472894	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
29	14.087736	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
30	16.196588	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
31	16.792386	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
32	16.911452	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
33	16.927112	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
34	17.884882	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
35	18.207872	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
36	18.222351	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
37	19.457516	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
38	21.460756	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
39	21.947420	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
40	22.084202	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
41	22.085661	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
42	22.089341	0c:ef:e8:27:00:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 0 Port = 0x8003
43	22.427874	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:85:91:42:00:00 Cost = 0 Port = 0x8001
44	22.442227	0c:85:91:42:00:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:85:91:42:00:00 Cost = 0 Port = 0x8001

Захват №3: Layer2Switch-2 (Backup, Eth7) <-> Layer2Switch-5 (Access, Eth3)

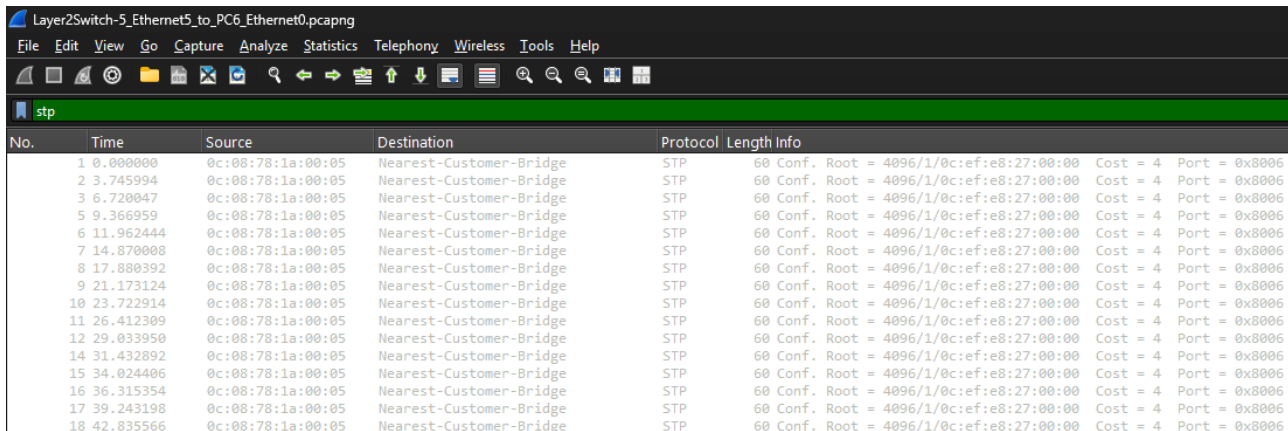
В пакете №1, отправленном со стороны Layer2Switch-2 (0ca9.cb37.0007), корневым коммутатором по-прежнему объявлен 0cef.e827.0000 (Layer2Switch-1), но транслируемая стоимость пути изменена на **Cost = 4**

Это доказывает, что некорневой коммутатор принимает BPDU на своем Root-порту, добавляет к нему стоимость этого корневого линка (в данном случае 4 для Gigabit-порта) и транслирует увеличенную стоимость дальше вниз к уровню доступа

Layer2Switch-2_Ethernet7_to_Layer2Switch-5_Ethernet3.pcapng						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
stp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
2	0.361492	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
4	0.544496	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
5	0.550103	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
6	1.392145	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
7	1.593007	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
8	1.705001	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
9	3.057853	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
10	4.705023	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
11	4.868194	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
12	4.870461	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
13	5.509374	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
15	6.650864	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
16	6.882520	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
17	6.981013	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
18	7.938867	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
19	8.723140	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
21	8.863884	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
22	8.866480	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
23	10.632612	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
24	11.666927	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
25	11.884378	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
26	12.031358	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
27	12.251643	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
28	12.347673	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
29	12.351644	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
30	13.595749	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
31	16.047403	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
32	16.213212	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
33	16.377449	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
34	16.397562	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
35	17.194357	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
36	17.456104	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
37	17.558020	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008
38	18.667741	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
39	20.862331	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
40	20.994264	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
41	21.006589	0c:08:78:1a:00:03	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:08:78:1a:00:00 Cost = 0 Port = 0x8004
42	21.431730	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8008
43	22.598416	0c:a9:cb:37:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:a9:cb:37:00:00 Cost = 0 Port = 0x8008

Захват №4: Layer2Switch-5 (Access, Eth5) <-> PC6

На данном линке фиксируются только исходящие BPDU-сообщения от коммутатора Layer2Switch-5 (0c08.781a.0005) со значением стоимости Cost = 4 в сторону ПК



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
2	3.745994	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
3	6.720047	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
5	9.366959	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
6	11.962444	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
7	14.870008	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
8	17.880392	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
9	21.173124	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
10	23.722914	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
11	26.412309	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
12	29.033950	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
14	31.432892	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
15	34.024406	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
16	36.315354	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
17	39.243198	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006
18	42.835566	0c:08:78:1a:00:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:ef:e8:27:00:00 Cost = 4 Port = 0x8006

ПУНКТ 5

Для демонстрации динамической сходимости STP на коммутаторе доступа Layer2Switch-3 была увеличена стоимость пути на его текущем корневом порту Gi0/0 с дефолтных 4 до 100

Команды изменения стоимости на Layer2Switch-3:

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#interface Gi0/0
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 100
vIOS-L2-01(config-if)#end
*Sep  7 14:06:14.242: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
```

Сравнение вывода Spanning Tree до и после изменения стоимости:

До изменений:

```
vIOS-L2-01>show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0cef.e827.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0c85.9142.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface    Role Sts Cost    Prio.Nbr Type
-----
Gi0/0        Root FWD 4     128.1   Shr
Gi0/1        Altn BLK 4     128.2   Shr
Gi0/2        Altn BLK 4     128.3   Shr
Gi0/3        Altn BLK 4     128.4   Shr
Gi1/0        Desg FWD 4     128.5   Shr
Gi1/1        Desg FWD 4     128.6   Shr
```

Порт Gi0/0 был корневым (Root FWD), а избыточный параллельный порт Gi0/1 был заблокирован (Altn BLK)

После изменений:

```
vIOS-L2-01#show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 4097
```

```
Address 0cef.e827.0000
```

```
Cost 4
```

```
Port 2 (GigabitEthernet0/1)
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
```

```
Address 0c85.9142.0000
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time 15 sec
```

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Altn	BLK	100	128.1	Shr	
Gi0/1	Root	FWD	4	128.2	Shr	
Gi0/2	Altn	BLK	4	128.3	Shr	
Gi0/3	Altn	BLK	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	
Gi1/1	Desg	FWD	4	128.6	Shr	

После увеличения стоимости интерфейса Gi0/0 до 100 этот путь перестал быть оптимальным к корневому коммутатору. Протокол STP автоматически пересчитал дерево топологии:

1. Порт Gi0/0 был переведен в заблокированное состояние (Altn BLK 100)
2. Резервный параллельный порт Gi0/1 со стоимостью 4 автоматически разблокировался и стал новым корневым портом (Root FWD), сохранив общую стоимость пути от коммутатора до корня равной 4
3. Точка подключения к корню (Root ID Port) переключилась с Port 1 (Gi0/0) на Port 2 (Gi0/1)

ПУНКТ 6

Демонстрация вывода конфигурации маршрутизатор Layer2Switch-3

```
vIOS-L2-01#show running-config
Building configuration...

Current configuration : 5072 bytes
!
! Last configuration change at 14:06:14 UTC Mon Sep 3 2026
!
version 15.0
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname vIOS-L2-01
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
vtp domain CISCO-vIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
```

```
vlan internal allocation policy ascending
!
vlan 100
  name VLAN100
!
vlan 200,300
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
  media-type rj45
  negotiation auto
  spanning-tree vlan 1 cost 100
!
interface GigabitEthernet0/1
  media-type rj45
  negotiation auto
!
interface GigabitEthernet0/2
  media-type rj45
  negotiation auto
!
interface GigabitEthernet0/3
  media-type rj45
  negotiation auto
!
interface GigabitEthernet1/0
  media-type rj45
  negotiation auto
!
interface GigabitEthernet1/1
  media-type rj45
```

```
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec ^C
*****
*****
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*****^C
banner incoming ^C
*****
*****
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*****
*****^C
banner login ^C
*****
*****
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*****^C
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 logging synchronous
 login
line vty 5 15
 logging synchronous
 login

```

!

!

end

ПУНКТ 7

Команда перехода на RSTP на коммутаторах:

<code>spanning-tree mode rapid-pvst</code>
--

Основные отличия в работе:

Роли портов: Заблокированные порты Altn BLK получили явную роль Alternate (Alt) - альтернативный некорневой порт, готовый мгновенно включиться в работу при падении основного корневого порта (RP) без прохождения стадий задержки.

Скорость сходимости: Сходимость в RSTP происходит практически мгновенно благодаря механизму рукопожатий Proposal/Agreement, в то время как классический STP требует прохождения таймеров Listening и Learning (суммарно до 30–50 секунд).

Генерация BPDU: В RSTP каждый коммутатор самостоятельно генерирует и отправляет BPDUs каждые 2 секунды со своих назначенных портов, выступая в роли Keepalive, тогда как в классическом STP некорневые коммутаторы лишь ретранслировали Hello-сообщения, полученные от корня.